

An LLM Agent for the Selection, Validation and Interpretation of Statistical Methods

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1. Before you start

Twenty-two slides, planned at about 21 minutes. If time is short, compress slides 13 to 16 — each is a single figure with one sentence to land — and keep slides 2, 12 and 18 at full length. Those three carry the problem, the honest reading of the results, and the evidence that the measurement itself was checked.

What a professor listens for is not a large number. It is whether you know exactly what your evidence supports, and whether you say where it stops before being asked.

2. The problem statement, at three lengths

In one sentence

“Language models produce statistical analyses that are fluent and arithmetically correct but often invalid, because validity depends on how the data was collected — which the data itself does not contain.”

In thirty seconds

“Ask an AI to analyse a dataset and it will always give you an answer. It reads well and the arithmetic is right. But whether the method is valid depends on facts about the study design — are the observations independent, paired, ordered in time — and none of that is visible in the spreadsheet. So the model can apply a method the data does not support, invent a statistic, or answer when it should decline. This project tests whether giving the model an explicit decision procedure fixes that.”

In two minutes

- Open with the observation: an LLM always answers, and its answers are fluent and arithmetically correct.
- Name the gap: validity is decided by design facts — independence, pairing, temporal order, clustering — that live outside the data file.
- Give the three consequences: an invalid method with correct numbers; a fabricated statistic that reads as a result; no obligation to decline.
- Say why it matters: the people most likely to rely on an AI's analysis are the people least able to audit its method.

- State the question: does an explicit protocol make the model choose valid methods more reliably than tool access alone?

3. Slide by slide

Slide 1 · Title

Establish who you are and the single question. (about 0.5 min)

- Introduce yourself and the project title.
- Read the question on the slide: does an explicit decision protocol make an LLM choose valid statistical methods more reliably than tool access alone?

Land this: Keep this short. The next slide does the real work.

Transition: “Let me start with why this question matters.”

Slide 2 · Problem statement

The most important slide. Everything after it answers it. (about 2 min)

- State the problem in one breath: LLM analyses are fluent, arithmetically correct, and frequently invalid.
- Explain why: validity depends on how the data was collected — independence, pairing, time order, clustering — and those facts are not in the data file.
- Walk the three consequences left to right: an invalid method with correct arithmetic; a fabricated number that reads like a result; and no mechanism that obliges the model to decline.

Land this: “The data itself does not contain the facts that decide whether the method is valid.”

Transition: “Here is what that looks like on real data.”

Slide 3 · Motivating case

Make the abstract problem concrete with one dataset. (about 1.5 min)

- Seoul Bike Sharing: 8,760 rows, one per hour for a year.
- The question — does temperature relate to demand — looks like a textbook correlation or Poisson regression.
- Both assume independent observations. Adjacent hours share weather, commuting cycles and daylight, so they are not independent.
- Nothing in the file says so. The design card supplies that fact.

Land this: “The table looked perfect, and the correct answer was to decline.”

Transition: “So the question I tested is this.”

Slide 4 · Research question and hypotheses

Show exactly what is being tested and against what. (about 1 min)

- Read the research question.
- H1 selection, H2 abstention, H3 provenance.
- Stress that H1 compares C with B, not with A. B has the same tools and the same report format, so B is the honest control.

Land this: “If C beats B, it is the procedure that won — not better tools.”

Transition: “Before the results, the design decisions that make this testable.”

Slide 5 · Scope and design decisions

Justify the closed library and the absence of code execution. (about 1 min)

- Fourteen methods across five task families, plus abstention.
- Closed so it is scorable — an open library turns method choice into essay grading.
- No general code-execution tool: it would let the model bypass the policy and make the three systems incomparable.
- Unsupported designs are not ignored; they are tested as abstention cases.

Land this: “Declining is a decision, not a failure.”

Transition: “Here is how the three systems are built.”

Slide 6 · Architecture

Pre-empt the fairness question. (about 1 min)

- Three drivers, one shared core: tool registry, result store, report schema, trace log.
- Only the control flow differs between A, B and C.

Land this: “Baseline fairness is provable from the code layout, not asserted.”

Transition: “The difference is in how C orders its work.”

Slide 7 · The decision protocol

Show the ordering that is the whole intervention. (about 1.25 min)

- Seven steps: parse, validate design, enumerate, diagnose, select or abstain, execute, verify.
- Point at steps 2 and 5. Design validation gates everything downstream, so a clean normality test can never talk the system into a method the design rules out.
- B can run every one of these checks too. What it lacks is the requirement to validate first, and the permission to stop.

Land this: “The intervention is the order, not the tools.”

Transition: “One more design decision, about numbers.”

Slide 8 · The provenance contract

Explain how fabrication is prevented rather than detected. (about 1 min)

- Every tool result is registered under a key.
- The report format rejects raw numbers in prose.
- The model writes a reference; references are substituted at the end; an unknown reference fails the run.
- Say the caveat yourself: for C this makes fidelity a guarantee by design, not an empirical finding. The empirical measure is the rejection rate.

Land this: “The model cannot write a number.”

Transition: “Now the test itself.”

Slide 9 · The benchmark

Establish that the answer key is not opinion. (about 1 min)

- 64 cases: 52 supported, 12 where declining is correct; 48 held out.
- Gold labels derive from the realised sample: design facts from the generator, distributional facts from the data actually drawn.
- Four public UCI datasets. The public labels were audited against their real values, which found and fixed one mislabel.

Land this: “48 of 64 labels are derived, not judged.”

Transition: “And the evaluation was designed to rule out tuning to the test.”

Slide 10 · Evaluation protocol

Show the controls against the most serious threat: leakage. (about 1 min)

- Held-out split frozen and never examined during development.
- Prompts fingerprinted by hash before the held-out run.
- Two repetitions per case, 288 graded runs, pinned model.
- Inference: per-case majority, then McNemar's exact test with a bootstrap interval — paired, because each case is answered by all three systems.

Land this: “Without the frozen split, a good result would only mean I had tuned to the answers.”

Transition: “So, the results.”

Slide 11 · Result — accuracy

Deliver the headline. (about 1 min)

- A 75%, B 67%, C 89%, with 95% intervals.
- C over B is +25 points against a +10 target.
- Point at B: tools without a procedure scored below no tools at all.

Land this: “Tools did not rescue the model. The ordering did.”

Transition: “Which of these differences are real?”

Slide 12 · Result — significance

Separate what is established from what is not. Do not rush this. (about 1.25 min)

- Solid and filled means significant; dashed and hollow means not.
- C over B: $p = 0.0018$. Established.
- C over A: +12.5 points, $p = 0.0703$. Not established — I report it as unresolved.
- B over A: not established either.

Land this: “A result I only believe when it agrees with me is not a result.”

Transition: “The sharpest separation is on the dangerous cases.”

Slide 13 · Result — abstention

The safety result. (about 0.75 min)

- On the design-hazard runs, C declined correctly 78% of the time.
- The tool-only baseline declined zero times — it fitted a real method to every invalid design.

Land this: “H2 is supported.”

Transition: “Then, could each system explain its result honestly?”

Slide 14 · Result — grounding

Show the second axis: reporting without invention. (about 0.75 min)

- The direct baseline had 43 of 96 reports rejected — a number no tool produced, or a check it never ran.
- B scores 100% here because it has tools and uses them. Its failure is choosing the wrong method, not inventing numbers.
- No fabricated number reached a reader in any system.

Land this: “H3 is supported.”

Transition: “Here is where each system actually goes wrong.”

Slide 15 · Result — failure stage

Show the mechanism behind every other result. (about 0.75 min)

- Total failures: A 24, B 32, C 11.
- Design-validation failures: B 20, C 4.

Land this: “The protocol removed exactly the failure it was built to remove.”

Transition: “It did give something up — coverage.”

Slide 16 · Result — risk and coverage

Answer the objection that C just refuses more. (about 0.75 min)

- C answered 85% of cases and was right on 87% of those.
- A and B answered 95% and 98%, and were right less often.
- Lower coverage came with higher accuracy on what was answered — the signature of declining the right cases, not declining at random.

Land this: “It declines when it should.”

Transition: “Pulling the findings together.”

Slide 17 · Hypotheses revisited

Summarise the findings against the questions asked. (about 0.75 min)

- H1, H2 and H3 supported.
- C over A unresolved — keep that row on the slide and say it.

Land this: “Three supported, one open.”

Transition: “Two things went wrong along the way, and they matter.”

Slide 18 · Measurement rigour

Demonstrate that you checked your own instrument. (about 1.25 min)

- Defect one: 51 runs ended in errors, but each had already chosen a method, and 37 had chosen correctly. Discarding them would have made the direct baseline look 6 points better than it is.
- Defect two: with two repetitions a tie had no majority, and the tie-break varied between sessions. The same data gave p-values from 0.016 to 0.125 — either side of 0.05. Fixed and enforced by a test.
- Neither raised an error. Both are documented as formal deviations.

Land this: “The system worked. The way I was measuring it did not, and nothing complained.”

Transition: “So here is what the evidence does not support.”

Slide 19 · Limitations

State the limits calmly and without apology. (about 1 min)

- Held-out claims are scoped to Claude Haiku 4.5.
- 48 cases and two repetitions resolve the main comparison but not C over A.
- The interpretation score is withheld: the blinded round failed its reliability check, kappa -0.044 .
- No independent second reviewer for the public labels; several design families are out of scope.

Land this: “Each limit is known, bounded and disclosed.”

Transition: “What the project contributes.”

Slide 20 · Contributions

Close the argument. (about 0.75 min)

- Controlled evidence that the protocol, not tool access, drives valid selection.
- A benchmark that scores declining.
- A report format in which an unsourced number cannot exist.

Land this: “Structure, not tooling, makes an LLM a reliable statistician.” — then pause.

Transition: “Where this goes next.”

Slide 21 · Future work

Show the obvious next experiments. (about 0.5 min)

- A larger held-out set to resolve C over A; frontier models; a second blinded scoring round; a wider method library.

Land this: “The design already tells me how much larger the sample needs to be.”

Transition: “You can see all of it running.”

Slide 22 · Live artefacts and questions

Invite scrutiny. (about 0.5 min)

- Point to the live explorer and the repository.
- If there is time, open the Seoul bike case and show C declining while A and B both fit a model.

Land this: “Thank you — I’m happy to take questions.”

4. Numbers to know without looking

Every figure below matches the recorded held-out run. Quote them exactly; rounding a p-value in speech is where a sharp question starts.

Measure	Value
Selection accuracy	A 75.0% · B 66.7% · C 88.5%
C over B	+25.0 points · $p = 0.0018$ · 95% CI [+12.5, +39.6]
C over A	+12.5 points · $p = 0.0703$ · not established
Declined correctly	A 28% · B 0% · C 78% (design-hazard runs)
Reports rejected	A 43 of 96 · B 0 · C 8
Design-validation failures	B 20 · C 4
Selective accuracy	C 87% at 85% coverage
Benchmark	64 cases · 52 supported · 12 abstain · 48 held out
Runs	288 graded, 2 repetitions, Claude Haiku 4.5
Reliability failure	inter-rater kappa -0.044

5. Questions to expect

Answer the question asked, then stop. Where a limit is involved, state it plainly — defensiveness reads worse than the limit itself.

Q. If fabrication is impossible in System C, isn't its perfect fidelity a tautology?

A. Yes, and that is how it is reported. For C, fidelity is a property of the design, not a measured finding. The empirical measure is how often the model reached for a reference that did not exist, and that rate is reported.

Q. Why is the main comparison C against B rather than C against A?

A. B holds exactly the same tools and the same report format as C; only the control flow differs. Comparing C with B isolates the protocol. Comparing C with A would mix the protocol with tool access.

Q. C over A isn't significant. Is the protocol actually better than no tools?

A. The direction favours C — the discordant cases split seven to one — but at 48 cases that is not enough to rule out chance, so I report it as unresolved. The primary comparison, against B, is significant at $p = 0.0018$.

Q. How do you know your gold labels are correct?

A. 48 of 64 labels are construction-derived: design facts come from the generator and distributional facts from the sample actually drawn, so they are not judgements. The 16 public-data labels were audited against their realised values, which caught and fixed one mislabel.

Q. Why only two repetitions?

A. It was a resource constraint. Two repetitions are enough to establish the primary comparison and to measure run-to-run stability — 25 of 144 system-case cells disagreed between runs — but not to characterise the full distribution. More repetitions are the first item of future work.

Q. Why Claude Haiku 4.5 and not a frontier model?

A. Every held-out claim is scoped to Haiku 4.5, and I do not generalise beyond it. The runner is model-agnostic and the evaluation is frozen, so re-running on a larger model is a direct next step.

Q. Why didn't you set temperature to zero for determinism?

A. The temperature parameter has been removed on current models and returns an error if sent. Determinism is controlled instead by a pinned model, a fixed effort level and a frozen prompt hash — and residual variance is measured across repetitions rather than assumed away.

Q. The expert policy scores 100%. Doesn't that prove the system works?

A. No — that score is circular, because the expert policy is what derived the gold labels. It validates the harness, not the method. The informative row is the naive policy inside the same state machine, which scores far lower and never abstains — showing the architecture alone does not produce the gain.

Q. Why did you keep errored runs in the accuracy figure?

A. Because none of the 51 errors were infrastructure. Each run had already chosen a method before its report failed the provenance check, and 37 had chosen correctly. Discarding them would raise the direct baseline from 75.0% to 81.1% by hiding its worst behaviour. Runs that never reached a decision are excluded.

Q. Couldn't System B simply do what System C does?

A. It has every tool C has, so nothing stops it. What it lacks is the requirement to validate the design before anything else, and the explicit permission to decline. The results show that without those, it does not discover them.

Q. What happened with the interpretation scoring?

A. The blinded round failed its reliability check — inter-rater kappa of -0.044 , worse than chance. The rubric admitted two defensible readings of its middle grade, so it was rebuilt. I withheld the metric rather than report a number from an instrument that demonstrably does not measure anything.

Q. Why McNemar's test?

A. Each case is answered by every system, so outcomes are paired and binary. McNemar's exact test uses only the discordant cases, which is right for paired binary data with small discordant counts. The bootstrap gives an interval on the difference itself.

Q. What is the practical application?

A. Anywhere an LLM analyses data for people who cannot audit the method — business analysts, clinicians, policy teams. The risk is not bad arithmetic; it is a confident answer from a method the data does not support.

Q. What would you do differently?

A. A larger held-out set and more repetitions to resolve C over A, and an independent second reviewer for the public-data labels from the start.

6. Phrases to avoid

Instead of	Say
“The system proves LLMs can be trusted.”	“On this model and benchmark, the protocol beat tool access alone by 25 points.”
“It is better than a plain LLM.”	“It leads the plain LLM by 12.5 points, but that difference is not significant at this sample size.”
“Numerical fidelity is 100%.”	“Fabrication is prevented by design; the rejection rate is the empirical measure.”
“It never makes mistakes.”	“It fails least at design validation, which is the failure it was built to prevent.”
“The results show it works on all models.”	“The held-out result is scoped to Claude Haiku 4.5.”